



Project Completed: Centralized Secret Management using AWS Secrets Manager & IAM

Project Overview

In this project, I eliminated hardcoded credentials from applications and implemented secure secret management using AWS Secrets Manager. The application dynamically retrieves secrets while access is controlled using IAM roles.

Key Steps

- Launched EC2 Instance
- Created Sample Application
- Demonstrated Hardcoded Credentials
- Created Secrets in AWS Secrets Manager
- Stored Database Username and Password
- Created IAM Role for EC2
- Configured Least Privilege Access
- Attached IAM Role to EC2 Instance
- Modified Application to Fetch Secrets Dynamically
- Removed Hardcoded Credentials
- Verified Secure Access to Secrets

Technologies Used

AWS EC2 | AWS Secrets Manager | IAM | Python | Security | DevSecOps

Architecture Components

- EC2 Instance
- AWS Secrets Manager
- IAM Role
- Secure Application
- Dynamic Secret Retrieval

EC2 Instance

The screenshot displays the AWS Management Console for the 'us-west-1' region. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, and Network & Security. The main content area shows the 'Instances (1/1)' page. A table lists the instance 'Project3' with its ID, state (Running), type (t3.micro), and other details. Below the table, the 'Details' tab for instance 'i-0e55b070070f0f779' is expanded, showing various attributes such as Instance ID, IPv6 address, Hostname type, Answer private resource DNS name, Auto-assigned IP address, Public IPv4 address, Instance state, Private IP DNS name, Instance type, VPC ID, Private IPv4 addresses, Public DNS, Elastic IP addresses, and AWS Compute Optimizer finding.

Secrets Manager Secret

The screenshot shows the AWS Management Console for the 'us-west-1' region, specifically the 'Secrets Manager' page. A green notification banner at the top indicates that a secret named 'project3_secret' has been successfully stored. Below the notification, the 'Secrets' section is displayed with a search bar and a table of secrets. The table has columns for 'Secret name', 'Description', 'Last retrieved (UTC)', and 'Created on (UTC)'. One secret is listed: 'project3_secret' with the description 'database credentials for project2', last retrieved at '-', and created on 'June 18, 2026 at 08:29:01'. A 'Store a new secret' button is visible in the top right corner.

IAM Role Configuration

The screenshot shows the AWS IAM console for the 'Ec2secretsRole'. The left sidebar contains the 'Identity and Access Management (IAM)' menu. The main content area displays the role's details, including a search bar, a filter by type dropdown, and a table of permissions. The table lists two permissions: 'AmazonSSMMangedInstanceCore' and 'SecretsManagerReadWrite', both of type 'AWS managed'. Below the table, there is a section for 'Permissions boundary (not set)' and a 'Generate policy based on CloudTrail events' button. The bottom of the console shows the 'CloudShell', 'Feedback', and 'Console Mobile App' links.

Security Group

The screenshot shows the AWS EC2 console for the 'Security Groups' page. The left sidebar contains the 'EC2' menu. The main content area displays a table of security groups. The table has columns for Name, Security group ID, Security group name, VPC ID, Description, and Owner. Two security groups are listed: 'default' and 'Project3'. The 'default' group has a description of 'default VPC security group' and the 'Project3' group has a description of 'Allows SSH,HTTP'. A 'Create security group' button is visible at the top right of the table.

Application Running Successfully

```
ec2-user@ip-172-31-0-24:~  
"username" : "admin",  
"password" : "admin123"  
}  
{'username': 'admin', 'password': 'admin123'}  
[ec2-user@ip-172-31-0-24 ~]$ python3 secret.py  
/home/ec2-user/.local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue receiving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/developer/python-support-policy-updates-for-aws-sdks-and-tools/  
  warnings.warn(warning, PythonDeprecationWarning)  
{  
"username" : "admin",  
"password" : "admin123"  
}  
{'username': 'admin', 'password': 'admin123'}  
[ec2-user@ip-172-31-0-24 ~]$ nano secret.py  
[ec2-user@ip-172-31-0-24 ~]$ python3 secret.py  
/home/ec2-user/.local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue receiving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/developer/python-support-policy-updates-for-aws-sdks-and-tools/  
  warnings.warn(warning, PythonDeprecationWarning)  
{  
"username" : "admin",  
"password" : "admin123"  
}  
{'username': 'admin', 'password': 'admin123'}  
{'username': 'admin', 'password': 'admin123'}  
admin  
admin123  
[ec2-user@ip-172-31-0-24 ~]$ |
```

Project Outcome

This project helped me understand Secret Management, IAM Security, Least Privilege Access, Credential Protection, and AWS Security Best Practices.

#AWS #DevSecOps #SecretsManager #IAM #CloudSecurity #Python #EC2